Category of Materials

- Metals
- Ceramics (compounds between metallic and non-metallic elements)
- Glasses (Amorphous Transparent solids)
- Polymers (Compounds composed of Carbon, hydrogen, and other non-metallic elements)
- Composites (Compounds of atleast 2 different material type
- Semiconductor

Crystaline → long range periodicity

Amorphous → NO periodicity

Silicon Carbon, Aluminium Nitride → Cermaic and Semiconductors

Property:

- External Effect ⟶ Response (in terms of State property)

Stress, Q, E — Strain, ΔT, ΔJ

Quotient of them is property

(Nothing to do with internal structure)

- Macroscopic Property: Bulk property, microscopic structure aint required

→ Electrical: Conductivity vs Electric field

→ Thermal: ΔT vs heat

→ Magnetic: Susceptibility vs magnetic field strength

→ mechanical: deformation vs stress, resistance to fracture
→ optical: refracted vs incident angle, attenuation of intensity
→ chemical: chemical reactivity, corrosion rate

Structure of material:

① Electronic Structure
② Atomic Structure
③ Nanostructure
④ Microstructure
⑤ Macrostructure

Electron microscopy and x-ray analysis → two most frequently used characterization techniques to see the micro/atomic/electronic structure.

4 important crieteria in material selections:

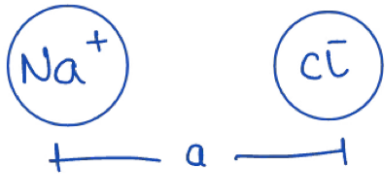
① in-service conditions to which the material will be subjected
② any deterioation of material properties during operation
③ economics / cost of fabricated piece
④ Toxicity or harmful nature of materials.

Thermal Expansion:

Thermal property 'α'.

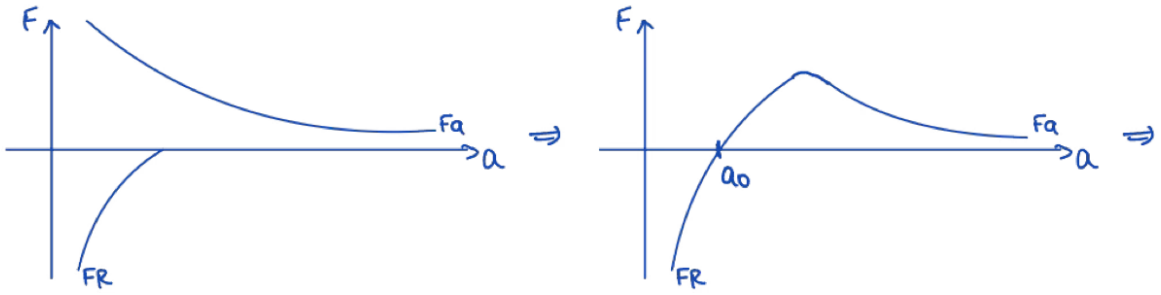
at T_0, length L_0

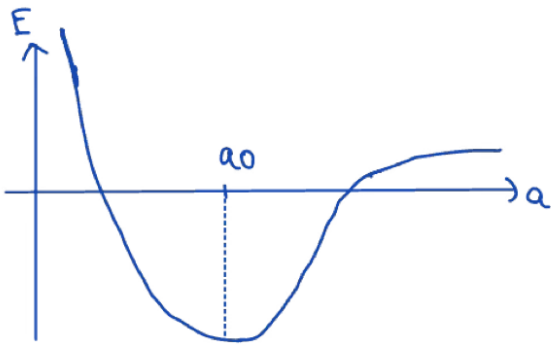
at T_1, length L_1 $\quad L_1 = L_0 + L_0\alpha(T_1 - T_0)$



$F_{attractive} = \frac{q^2}{4\pi\varepsilon_0 a^2}$

$F_{repulsive} = \lambda e^{-a/\ell}$ $\quad -F = F_A + F_R$

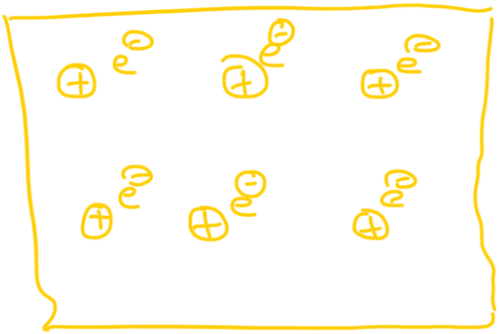




as Temperature increases, Energy decreases

Electrical Conductivity:

Free electron gas:



consider SCC - structure

$\Rightarrow r = \frac{a}{2}$ (a → cube side)

$P.F = \frac{4/3\pi(a/2)^3 \times 1}{a^3} = \pi/6 \sim 50\%$

$\Rightarrow$ 50% of the cell is vacant and also e's are

→ 50% of the cell is vacant and ... is air

much smaller than atoms. $e^{\ominus}$ can be treated as gas particles (bound to the cell though)

Number of gas atoms (assume monovalent) per cc

[1STP = 1mol of gas = 22.4L; 6.023×10^{23}]

$\Rightarrow 3 \times 10^{19}$ atoms/cc

Ag : $\rho = 10.5$ g/cc 107 amu valency = +1

$\Rightarrow$ atoms/cc $= \frac{10.5}{107} \times 6.023 \times 10^{23}$ atoms/cc

$= 6 \times 10^{22}$ atoms/cc

$= 6 \times 10^{22}$ $e^{\ominus}$'s/cc

- The above model explains electrical & thermal conductivity.

Pro's:

① electrons undergo collisions which are instantaneous and result in scattering

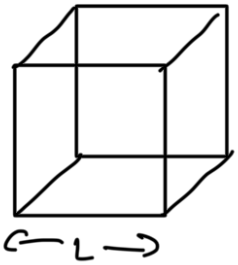
② In between two collisions, all interactions are neglected; but an average resistive term is incorporated to take care of the effects.

③ mean time between collision of electrons is independent of position and velocity.

④ Electrons obtain an equilibrium with the

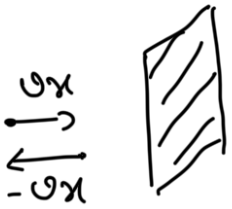
surroundings based on collision with other electrons.

Kinetic Theory of Gases:



n → moles

m → mass



$\Delta P = -2mv_x$

$\Rightarrow \Delta P$ (surface) $= 2mv_x$

Time between 2 collisions $= \frac{2L}{v_x}$

$\Rightarrow \frac{\Delta P}{\Delta t} = \frac{2mv_x}{2L/v_x} = \left(\frac{m}{L}\right)v_x^2 = F$ (by 1 particle)

total force $= \frac{m}{L}\left(\sum_{i=1}^{N} v_{x_i}^2\right)$ $\qquad N = n.N_A$

$v^2 = v_x^2 + v_y^2 + v_z^2$ (equally probable) $= 3v_x^2$

$\Rightarrow v_x^2 = v^2/3 \Rightarrow \frac{m}{L}.N.\langle v_x^2\rangle = \frac{m.n.N_A}{L}\langle v_x^2\rangle$

$P = \frac{F}{A} = \frac{mnN_A}{L^3}\langle v_x^2\rangle$ $\qquad L^3 = V$

$\Rightarrow PV = \frac{mnN_A\langle v^2\rangle}{3} = nRT$

$\Rightarrow m\langle v^2\rangle = 3k_BT$

Avg KE per molecule $= \frac{3}{2}k_B.T$

Total KE / volume $= \frac{3}{2} K_B N_T . T \rightarrow$ Thermal Energy per vol

Specific heat capacity : $\frac{dE}{dt} = \frac{3}{2} N_T K_B = C_V^e$ (ions → excluded)

$V = iR = i\frac{\rho L}{A} \Rightarrow J = \sigma E$ $\quad E \rightarrow$ Electric Field

$J = \underbrace{e . n}_{\text{known}} v$ (unit cross section area) $\quad n \rightarrow$ number density

$v = \frac{p(t)}{m}$ $\quad p(t) \rightarrow$ momentum of $e^{\ominus}$

($\tau \rightarrow$ mean free time)

$p(t+\Delta t) = ?$

fraction of $e^{\ominus}$ that wouldn't collide $= 1 - \frac{\Delta t}{\tau}$

$$\left(1 - \frac{\Delta t}{\tau}\right)\left\{ p(t) + f(t)\,dt \right\} + \frac{\Delta t}{\tau} K f(t)\,dt = p(t+\Delta t)$$

$$\Rightarrow p(t+\Delta t) - p(t) = f(t)\,dt - \frac{\Delta t}{\tau} f(t)\,dt + \frac{\Delta t}{\tau} K f(t)\,dt - \frac{\Delta t}{\tau} p(t)$$

$$p(t+\Delta t) - p(t) = f(t)\,\Delta t - p(t)\frac{\Delta t}{\tau} - \frac{\Delta t}{\tau}\left[f(t) - K f(t) \right] dt$$

$$\Rightarrow \boxed{\frac{d}{dt} p(t) = f(t) - \frac{p(t)}{\tau}} \qquad (dt \rightarrow 0)$$

$$m\frac{dv}{dt} = e\bar{E} - \beta v$$

At steady state : $m dv = 0 \Rightarrow e\bar{E} - \beta v_d = 0$

$$\frac{dp}{dt}$$

v_f = drift velocity $\quad m\frac{dv}{dt} = e\bar{E}\left(1-\frac{v}{v_f}\right)$

$$\Rightarrow \frac{dv}{1-v/v_f} = \frac{e\bar{E}}{m}dt \Rightarrow -v_f \ln\left[1-\frac{v}{v_f}\right] = \frac{e\bar{E}}{m}t + c$$

$$t=0;\ v=0 \Rightarrow c=0 \Rightarrow -v_f \ln\left[1-\frac{v}{v_f}\right] = \frac{e\bar{E}t}{m}$$

$$\Rightarrow \boxed{v = v_f\left\{1-e^{-\frac{e\bar{E}t}{mv_f}}\right\}}$$

also $v = u+at$

$$\Rightarrow v_f = 0+\frac{e\bar{E}}{m}\tau$$

$$\Rightarrow \boxed{\frac{v_f m}{e\bar{E}} = \tau}$$

$$\Rightarrow \boxed{v = v_f\left\{1-e^{-t/\tau}\right\}}$$

$$J = env = eN_f v_f = eN_f \cdot \frac{e\bar{E}\tau}{m} \Rightarrow \boxed{\sigma = \frac{e^2 N_f \tau}{m}}$$

→ $e^{\ominus}$s have scattering time τ, the probability of scattering within a time interval $dt = dt/\tau$

→ once scattering occurs, we assume $e^{\ominus}$s returns to momentum $p=0$

→ b/w scattering events, $e^{\ominus}$s respond to external electric and magnetic fields

$e^{\ominus}$ have momentum p at time t

$$\langle p(t+dt)\rangle = \left(1-\frac{dt}{\tau}\right)\left(P(t)+Fdt\right) + 0\cdot dt/\tau$$

$$dP/dt = F - P/\tau$$

drag force kinda

For metals:

① Free electron gas → 4 rules

② KTG
- ↳ Avg KE per molecule: $3/2\ K_B T$
- ↳ $C_V = 3/2\ N_f K_B$

③ Ohm's law: $V = IR$ (or) $J = \sigma E$ — σ: electrical conductivity

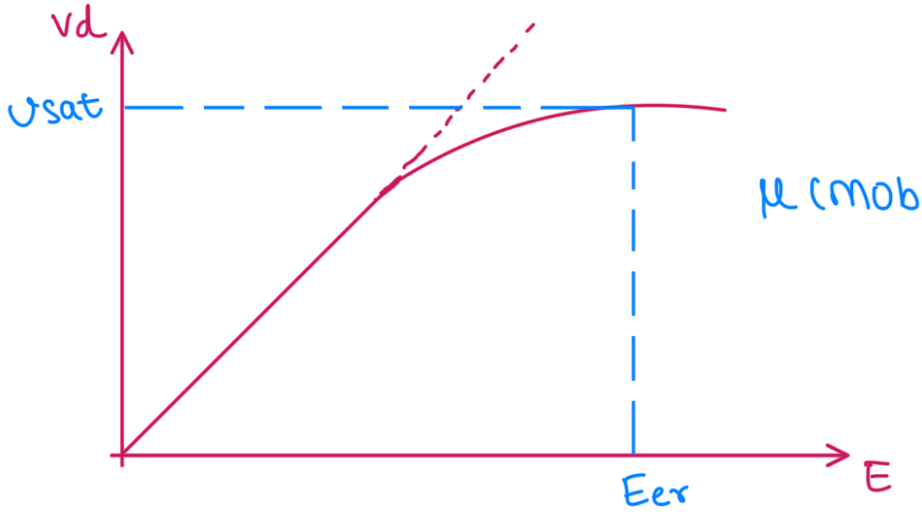
④ Drude-Lorentz model:

Electric conductivity $\sigma = \frac{N_f e^2 \tau}{m}$

⑤ $\frac{dp(t)}{dt} = f(t) - \frac{p(t)}{\tau}$ ↳ average resistive term

⑥ $J = ne v_d$ v_d → drift velocity

v_d is in the order of $10^{-2} - 10^{-4}$ m/s



μ (mobility) $= \frac{v_d}{E}$ (slope)

Heat Transport

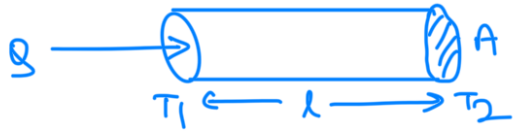
metals

→ radiation (EM waves)

→ conduction ⇄ collision of electrons ↺
⇄ atomic vibrations

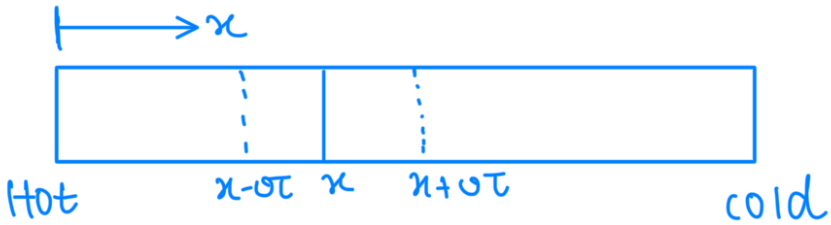
→ convection (actual transport of atoms)
(liquids & gases)

Thermal conductivity: K



$$K = -\frac{\frac{Q}{tA}}{\Delta T/\Delta x}$$

J_Q (Heat flux) $= -K\frac{dT}{dx}$ fourier's law of heat conduction



Thermal Energy E is a function of temperature and temperature is a function of x.

$v \rightarrow$ mean velocity

$\tau \rightarrow$ mean time between collisions

($\frac{1}{2}$ since 1D motion, $\frac{1}{2} \rightarrow$ left, $\frac{1}{2} \rightarrow$ right)

$$J_{Hot\ to\ cold} = \frac{Nf}{2} \times E(T(x - v\tau)) \cdot v$$

$$J_{cold\ to\ hot} = \frac{Nf}{2} \times E(T(x + v\tau)) \cdot v$$

$$J_{net} = \frac{Nfv}{2}\{E(T(x - v\tau)) - E(T(x + v\tau))\}$$

$E(T(x)) - v\tau\frac{dE}{dx}$ $\quad$ $E(T(x)) + v\tau\frac{dE}{dx}$

$$J_{net} = \frac{N_f v}{2}\left(-2v\tau\frac{dE}{dx}\right) = N_f v^2 \tau\left(-\frac{dE}{dx}\right)$$

$$\Rightarrow J_{net} = N_f v^2 \tau \left(-\frac{dE}{dT}\right)\left(\frac{dT}{dx}\right)$$

$\left(-\frac{dE}{dT}\right) \rightarrow C_v^e$

$$\Rightarrow J_{net} = -C_v^e v^2 \frac{dT}{dx}$$

considering all directions now.

$$J_{net} = -\frac{1}{3}\langle v^2\rangle \tau C_v^e \frac{dT}{dx}$$

$$\Rightarrow K = \frac{1}{3}\langle v^2\rangle \tau C_v^e$$

$m/s \quad s \quad J/m^3K \Rightarrow Wm^{-1}K^{-1}$

$C_v^e = 3/2\, n k_B$ $n \rightarrow$ no. density
$v \rightarrow$ rms

$v \sim 10^5 m/s \quad \tau \sim 10^{-14} s \quad C_v^e \sim 10^6 J m^{-3} K^{-1} \Rightarrow K \sim 10^2 Wm^{-1}K^{-1}$

Wiedemann-Franz law: Electrical and Thermal conductivity are correlated.

$$\frac{K}{\sigma T} \sim 2.4 \times 10^{-8}\ J\Omega s^{-1} K^{-2}$$

$K = 1/3 \langle v^2\rangle \tau C_v^e$ from drude's model

$\sigma = \frac{N_f e^2 \tau}{m}$ from drude's model

$$\Rightarrow \frac{K}{\sigma T} = \frac{1}{3T}\frac{m\langle v^2\rangle C_v^e}{N_f e^2}$$

$m v^2 = 3 k_B T$

$C_v^e = 3/2 N_f k_B$

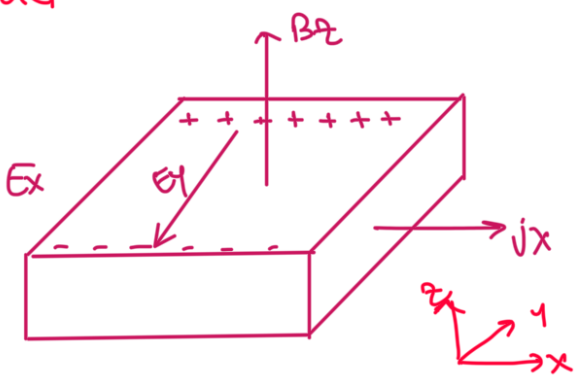
$$\frac{K}{\sigma T} = \frac{k_B^2}{e^2} = 1.1 \times 10^{-8}\ J\Omega s^{-1} K^{-2}$$

(very close $\Rightarrow$ good model)

limitations of drude's model

disadvantages of drudes model

① Hall effect:



$$\frac{dp}{dt} = -(eE + e(\bar{v} \times \bar{B})) - \frac{p}{\tau}$$

$$\frac{dp}{dt} = -e[\bar{E} + \bar{v} \times \bar{B}] - \frac{p}{\tau} \qquad \bar{j}_x = -ne\bar{v} \qquad p = mv$$

under steady state; $\frac{dp}{dt} = 0$

$$0 = -e\left[\bar{E} - \frac{\bar{j}_x}{ne} \times \bar{B}\right] + \frac{m\bar{j}_x}{ne\tau}$$

$$\Rightarrow \bar{E} = \frac{1}{ne}(\bar{j}_x \times \bar{B}) + \frac{m\bar{j}_x}{ne^2\tau}$$

define hall coefficient

$R_H = \frac{E_y \rightarrow \text{due to hall effect}}{j_x \times B_z}$

$E_y = -\frac{j \cdot B_z}{ne}$ (1st term basically) $\Rightarrow$ $\boxed{R_H = -\frac{1}{ne}}$

drude's can't predict -ve hall coefficient

② Thermoelectric effect:

j^q (thermal current) $= \Pi\, j^e$ (electric current)

$\Pi \rightarrow$ peltier coefficient

$j^q = \frac{1}{2} K_B T \cdot n \cdot v_x$ (each $e^\ominus$) $\rightarrow$ 100 times overestimated

$j^e = -nev_x \Rightarrow \Pi = -\frac{1}{2}\frac{K_B T}{e}$

$\hookrightarrow$ 100 times overestimated

Seibeck coefficient $= \frac{\Pi}{T} = -\frac{1}{2}\frac{K_B}{e} = -4.3 \times 10^{-5}\ V/K$

③ drude's model overestimates c_v^e and underestimate

velocity hence good k approximation